

7 (a) (i) State what is meant by isotopes.

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.....

..... [2]

- (ii) A thorium-232 nucleus ( $^{232}_{90}\text{Th}$ ) undergoes radioactive decay to form another nucleus. A sequence of further decays occurs until a nucleus is formed that is another isotope of thorium. In each decay, either a single  $\alpha$ -particle or a single  $\beta^-$  particle is emitted.

Determine the minimum numbers of  $\alpha$ -particles and  $\beta^-$  particles that could be emitted during the sequence of decays to form the other isotope of thorium from thorium-232.

minimum number of  $\alpha$ -particles = .....

minimum number of  $\beta^-$  particles = .....

[2]

- (iii) Complete Figure 7.1 to show the nucleon number and proton number of the thorium isotope formed by the sequence of decays.

.....  
..... Th

**Figure 7.1**

[2]

- (b) (i) State the quark composition of a proton.

..... [1]

- (ii) Describe the change to the quark composition of the proton when it undergoes  $\beta^+$  decay.

.....

..... [1]

[Total: 8]